



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium



How the System Works

Inputs

Comprehensive Data Collection

The system continuously ingests three primary categories of data to build a 360-degree view of customer risk:

- **Behavioral Data:** Recent repayment history (6-month trends), transaction frequency, and patterns of late vs. on-time payments.
- **Financial Health Metrics:** Current credit utilization ratios, total loan balances, and Debt-to-Income (DTI) calculations.
- **Contextual Attributes:** Foundational data including Credit Scores, employment status, and account tenure.

How the System Works

Decision Logic

The Hybrid Intelligence Brain

Decisions are not based on a single factor but a combination of two powerful logic streams:

- **Predictive Modeling (Random Forest):** An ensemble machine learning model calculates a "Risk Probability Score" for every customer. It identifies non-linear patterns, such as a customer whose risk spikes only after utilization crosses 80%.
- **Automated Business Rules:** Hard-coded "Guardrails" for immediate action. For example, any account exceeding 100% utilization (Over-Limit) is immediately flagged for a manual review regardless of their predicted score.

How the System Works

Actions

Tiered Intervention & Outreach

Based on the Risk Score and Decision Logic, the system triggers specific, automated workflows:

- **Low/Standard Risk:** Automated "Nudges" via SMS or Push notifications, providing financial literacy tips or upcoming payment reminders.
- **High/Emerging Risk (The "Hidden Risk" Segment):** Initiation of the "Credit Wellness" protocol. This includes personalized, GenAI-drafted offers for temporary interest rate reductions or flexible, restructured payment schedules.
- **Critical Risk:** Direct escalation to the specialized Collections team for high-touch, human intervention and hardship negotiation.

How the System Works

Learning

The Optimization Loop

The system is designed to get smarter with every customer interaction:

- **Outcome Tracking:** Every action (e.g., an SMS sent or a rate reduction offered) is tracked against the eventual outcome (did the customer pay?).
- **Automated Feedback Loop:** This success/failure data is fed back into the training set.
- **Model Retraining:** The Random Forest model is periodically retrained on the most recent 30 days of data, allowing it to adapt to changing economic conditions or new customer spending trends.

Role of Agentic AI

Autonomous (Agent-Led)	Human Oversight (Human-in-the-Loop)
Real-Time Risk Monitoring: Using SQL RAG to autonomously identify utilization spikes and flag the "Hidden 89" segment as soon as they meet risk thresholds.	Strategic Approval (HITL): Reviewing and authorizing high-impact hardship offers or debt restructuring plans drafted by the AI before they are sent.
Policy & Compliance Audit: Using Vector RAG to scan internal policy PDFs and regional regulations to ensure every outreach attempt is 100% compliant.	Complex Case Management: Handling unique customer edge cases or emotional negotiations that require human empathy and nuanced judgment.
Hyper-Personalized Drafting: Utilizing Prompt Engineering to generate empathetic, context-aware messages tailored to a customer's specific financial history.	System Health & Safety (HOTL): Monitoring Guardrails to prevent biased outcomes and managing Model Drift through MLOps to ensure long-term accuracy.
Scalable "Nudges": Automatically deploying low-stakes payment reminders and "Financial Wellness" tips via SMS or app notifications.	Final Accountability: Serving as the ultimate authority for decisions that significantly affect a customer's legal standing or credit bureau reporting.

Responsible AI Guardrails

To ensure Geldium's AI-powered collections system operates with the highest ethical standards, we implement a multi-layered safety framework.

- **Fairness & Bias Mitigation: Continuous Monitoring:** Regular "Disparate Impact" testing to ensure risk scores don't unfairly penalize specific demographics (Age, Location, Employment Type).
 - **Feature Neutralization:** Systematically removing "proxy variables" that could lead to indirect discrimination, ensuring decisions are based purely on financial behavior.
- **Explainability & Transparency (SHAP Integration):**
 - **The "Why" Behind the Score:** Every AI-generated risk score is accompanied by a SHAP value breakdown. This translates complex model weights into plain-English reasons (e.g., *"Flagged due to a 30% increase in DTI ratio"*).
 - **Stakeholder Clarity:** Ensures the Collections team can clearly communicate the rationale for any intervention to the customer.

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- **Regulatory Compliance & Auditability:**
 - **Compliance-by-Design:** The system's **Vector RAG** component cross-references all outreach scripts against standards like **ECOA** (Equal Credit Opportunity Act) and local banking regulations.
 - **Immutable Logs:** Every decision, automated nudge, and hardship offer is logged in a tamper-proof audit trail for regulatory reviews.
- **Human-in-the-Loop (HITL) Safeguards:**
 - **Critical Decision Gating:** No punitive actions or complex legal restructuring can be finalized without a manual review and sign-off from a senior collections specialist.
 - **Kill-Switch Protocol:** A manual override capability to pause the system instantly if any "Model Drift" or anomalous pattern is detected in the learning loop.

PII Protection & Data Security Policy

To ensure all predictive activities and agentic automations adhere to the highest standards of data privacy, maintaining compliance with global and local regulations (GDPR, ECOA).

- **Identity De-Identification & Tokenization:**

- **The "Vault" Approach:** Real-world identifiers (Names, Emails, SSNs) are stored in a secure, encrypted vault. The AI engine only processes **Anonymized Tokens** (e.g., User_ID_8921) to perform risk assessments.
- **Data Minimization:** The AI is only granted access to behavioral data (utilization, payment history) required for the prediction, excluding sensitive demographic PII unless strictly necessary for compliance.

- **Privacy-First Agentic Architecture:**

- **Automated Redaction:** An "Intermediate Scrubbing Layer" automatically redacts PII from any customer query or notes before they reach the Large Language Model (LLM).
- **Zero-Data Retention:** We utilize "Enterprise-Grade" API agreements where customer data is processed in real-time but **never stored** or used to train the provider's general AI models.

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- **Access Control & Accountability:**

- **Role-Based Access (RBAC):** Only human "Financial Wellness Counselors" have the authorization to re-identify a customer once a risk flag is triggered, ensuring a clear audit trail.
- **Auditability:** Every interaction where the agent accesses a customer profile is logged in an immutable ledger, providing 100% transparency for regulatory audits.

- **Model Security:**

- **Differential Privacy:** Injecting mathematical "noise" into the training set to ensure the model learns *patterns* (e.g., "High DTI = Risk") without being able to memorize or reveal individual customer balances.
- **Regular PII Audits:** Bi-monthly scans of the Vector Database to ensure no unstructured customer data has accidentally leaked into the "semantic memory" of the system.

Expected Business Impact

The implementation of an autonomous, proactive collections system shifts Geldium Finance from "Loss Mitigation" to "Risk Prevention," delivering measurable value across the entire organization.

Business KPIs (Quantitative Outcomes)

- **15% Reduction in DPD Transition:** By targeting the "Hidden Risk" segment (such as the 89 customers identified) before their first missed payment, we aim to significantly lower the volume of accounts moving into 30+ Days Past Due status.
- **Increased Net Repayment Rates:** Proactive intervention via flexible hardship offers is projected to increase total recovery on at-risk balances by 20% compared to traditional, reactive collections.
- **Operational Cost Efficiency:** Utilizing a Multi-Agent system for monitoring and initial outreach reduces the "cost-to-collect" by 40%, allowing human specialists to focus exclusively on high-complexity negotiations.
- **Lower Capital Reserve Requirements:** Reducing the Non-Performing Loan (NPL) ratio allows Geldium to release capital currently held in reserve, freeing up liquidity for new lending products.

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Customer & Operational Outcomes (Qualitative Results)

- **Improved Customer Trust:** Shifting the narrative from "Debt Collection" to "Financial Wellness" preserves the customer relationship and increases long-term loyalty.
- **Enhanced Fairness & Inclusion:** By replacing human intuition with auditable, behavioral-based AI scoring (SHAP), we eliminate unconscious bias and ensure every customer is treated equitably.
- **Global Scalability:** The autonomous nature of the agentic system allows Geldium to manage a 10x increase in account volume without a linear increase in headcount.
- **Regulatory Resilience:** With an immutable audit trail and built-in compliance guardrails, Geldium is better prepared for audits and changing financial regulations (e.g., ECOA).

Conclusion & Strategic Outlook

Transforming Collections into a Competitive Advantage

- **From Reactive to Proactive:** By leveraging the 89 "**Hidden Risk**" customer insights, we move from chasing debt to preserving capital. We are now ahead of the delinquency curve, not behind it.
- **Trust Through Transparency:** The integration of **SHAP Explainability** and our **PII Vault** ensures that Geldium leads the market in Ethical AI. We are prepared for future audits and regulatory shifts (ECOA/GDPR).
- **Scalable Intelligence:** The **Multi-Agent architecture** (Hybrid RAG) allows the bank to manage portfolio growth without an exponential increase in operational overhead.
- **The Bottom Line:** This system protects Geldium's balance sheet while enhancing the financial wellness of our customers—creating a more resilient and responsible institution.